3D Matrices: Eigenvalues and Eigenvectors

Questions 1–18

Question 1. Show that each of the following vectors are eigenvectors of the given matrix $\mathbf{A}$ and in each case state the associated eigenvalue:

a) $\begin{pmatrix}1\\1\\0\end{pmatrix}$, $\mathbf{A} = \begin{pmatrix}2 & 1 & 0\\1 & 2 & 0\\0 & 0 & 4\end{pmatrix}$

b) $\begin{pmatrix}0\\1\\2\end{pmatrix}$, $\mathbf{A} = \begin{pmatrix}1 & -2 & 1\\1 & 1 & 1\\1 & 2 & 2\end{pmatrix}$

c) $\begin{pmatrix}1\\3\\2\end{pmatrix}$, $\mathbf{A} = \begin{pmatrix}-7 & 1 & 0\\-16 & 0 & 2\\-7 & -1 & 1\end{pmatrix}$

Question 2. Given the matrix

$$\mathbf{M} = \begin{pmatrix}-11 & -6 & 10\\-4 & -9 & 8\\-10 & -12 & 14\end{pmatrix}$$

a) Show that the characteristic equation of $\mathbf{M}$ is given by $(\lambda - 3)(\lambda + 3)(\lambda + 6) = 0$ where λ is an eigenvalue of $\mathbf{M}$

b) Hence state the eigenvalues of $\mathbf{M}$

Question 3. Find the eigenvalues of each of the following matrices:

a)

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & -3 \end{pmatrix}$$

b)

$$\begin{pmatrix} 1 & -2 & 0 \\ 2 & 1 & 5 \\ 0 & 0 & 5 \end{pmatrix}$$

c)

$$\begin{pmatrix} 0 & -1 & 1 \\ -1 & -1 & 2 \\ -1 & 0 & 1 \end{pmatrix}$$

d)

$$\begin{pmatrix} 1 & -1 & -3 \\ 0 & 4 & -8 \\ 0 & 4 & -4 \end{pmatrix}$$

Question 4. Show that the matrix

$$\mathbf{A} = \begin{pmatrix} p & -q & 0 \\ q & p & 0 \\ 0 & 0 & r \end{pmatrix}$$

for constants p, q and r where $q \neq 0$ has complex eigenvalues $\lambda = p \pm q\mathrm{i}$ and $\lambda = r$

Question 5. Given that the matrix

$$\mathbf{A} = \frac{1}{2}\begin{pmatrix} 4 & 0 & -3 \\ 0 & -10 & 11 \\ 0 & 0 & 1 \end{pmatrix}$$

a) Find the eigenvalues of $\mathbf{A}$
b) By first finding the matrix $\mathbf{A}^2$ calculate the eigenvalues of $\mathbf{A}^2$
c) Determine the eigenvalues of $\mathbf{A}^2$ in terms of the eigenvalues of $\mathbf{A}$
d) Show that in general, $\mathbf{M}^2\mathbf{v} = \lambda^2\mathbf{v}$ where λ represents an eigenvalue of matrix $\mathbf{M}$

Question 6. Given that the matrix

$$\mathbf{A} = \begin{pmatrix} 3 & -3 & 1 \\ 0 & -5 & 6 \\ 0 & -3 & 4 \end{pmatrix}$$

a) Find the eigenvalues of $\mathbf{A}$
b) By first finding the matrix $\mathbf{A}^3$ calculate the eigenvalues of $\mathbf{A}^3$
c) Determine the eigenvalues of $\mathbf{A}^3$ in terms of the eigenvalues of $\mathbf{A}$
d) Show that in general, $\mathbf{M}^3\mathbf{v} = \lambda^3\mathbf{v}$ where λ represents an eigenvalue of matrix $\mathbf{M}$

Question 7. Given the matrix

$$\mathbf{M} = \begin{pmatrix} -5 & 2 & 2 \\ -18 & 7 & 6 \\ -6 & 2 & 3 \end{pmatrix}$$

a) Show that $\mathbf{M}^2 + 3\mathbf{I} = k\mathbf{M}$ where k is a constant to be found.
b) Hence show that

$$\mathbf{M}^{-1} = a\mathbf{M} + b\mathbf{I}$$

where a and b are constants to be found.
c) Find the eigenvalues of $\mathbf{M}$
d) Show that $\lambda^2 + 3 = k\lambda$ for each of your eigenvalues and your value of k

Question 8. Given the matrix

$$\mathbf{M} = \begin{pmatrix} 1 & 3 & -1 \\ -2 & 3 & 2 \\ 1 & 0 & 2 \end{pmatrix}$$

a) Show that the characteristic equation of $\mathbf{M}$ is given by $(\lambda - 3)(\lambda^2 - 3\lambda + 9) = 0$ where λ is an eigenvalue of $\mathbf{M}$
b) Hence find the eigenvalues and corresponding eigenvectors of $\mathbf{M}$

Question 9. Find the eigenvectors for the following matrices for the given eigenvalue:

a) $\begin{pmatrix} 6 & 3 & 6 \\ -2 & -3 & -10 \\ -8 & -3 & -4 \end{pmatrix}$, $\lambda = 2$

b) $\begin{pmatrix} -2 & 5 & 7 \\ 1 & 0 & -1 \\ -1 & 1 & 2 \end{pmatrix}$, $\lambda = 0$

c) $\begin{pmatrix} 4 & -8 & 3 \\ 3 & -7 & 3 \\ 4 & -10 & 5 \end{pmatrix}$, $\lambda = -1$

d) $\begin{pmatrix} -1 & 1 & 2 \\ 4 & 2 & -1 \\ -4 & 1 & 3 \end{pmatrix}$, $\lambda = 3$

e) $\begin{pmatrix} -5 & 1 & 1 \\ \frac{3}{2} & 0 & 2 \\ -1 & 3 & 0 \end{pmatrix}$, $\lambda = -5$

Question 10. Given the matrix

$$\mathbf{M} = \begin{pmatrix} 4 & 1 & 0 \\ 1 & 4 & 0 \\ 1 & 1 & 5 \end{pmatrix}$$

a) Show that 5 is a repeated eigenvalue of $\mathbf{M}$

b) State all of the eigenvalues of $\mathbf{M}$

c) For each eigenvalue, find the corresponding eigenvectors.

Question 11. Given the matrix

$$\mathbf{M} = \begin{pmatrix} 6 & 1 & -4 \\ 3 & 4 & -4 \\ 7 & 1 & -5 \end{pmatrix}$$

a) Show that 3 is an eigenvalue of $\mathbf{M}$
b) Find the other eigenvalues.
c) For each eigenvalue, find the corresponding eigenvectors.

Question 12. Given the matrix

$$\mathbf{M} = \begin{pmatrix} -4 & 16 & -8 \\ -5 & 13 & -6 \\ -10 & 22 & -10 \end{pmatrix}$$

a) Show that 2 is an eigenvalue of $\mathbf{M}$
b) Find the other eigenvalues.
c) For each eigenvalue, find the corresponding eigenvectors.

Question 13. Given the matrix

$$\mathbf{M} = \begin{pmatrix} -1 & p & 0 \\ p & 0 & 2 \\ 0 & 2 & 1 \end{pmatrix}$$

where p is a real constant.
One of the eigenvalues of matrix $\mathbf{M}$ is 3
The corresponding eigenvector is $\begin{pmatrix} 1 \\ 2 \\ q \end{pmatrix}$ for constant q
Find the value of p and q

Question 14. It is given that matrix

$$\mathbf{M} = \frac{1}{14}\begin{pmatrix} 10 & -44 & -2k \\ 0 & 42 & 0 \\ 6 & -53 & -17 \end{pmatrix}$$

where k is a positive integer.
It is known that the eigenvalue $\lambda_1 = 3$ has corresponding eigenvector $\mathbf{v}_1$
The other two eigenvectors are $\begin{pmatrix} a \\ 0 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$
where a is a positive integer.

a) Find the value of k
b) Determine the value of a
c) State the values of all the eigenvalues.

Question 15. Given the matrix

$$\mathbf{M} = \begin{pmatrix} 6 & -2 & 2 \\ 2-k & 2+k & k-2 \\ 4-k & k-4 & k+4 \end{pmatrix}$$

where k is a real constant. Two of the eigenvalues of matrix $\mathbf{M}$ are 4 and 8

a) Find the third eigenvalue of $\mathbf{M}$ giving your answer in terms of k

The eigenvectors of $\mathbf{M}$ corresponding to the eigenvalues 4 and 8 are $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ respectively.

b) Find the eigenvector corresponding to the third eigenvalue.

Question 16. Given the matrix

$$\mathbf{M} = \begin{pmatrix} k-2 & -3 & 3 \\ 6 & -5 & 3 \\ 0 & 0 & -2 \end{pmatrix}$$

where k is an integer,

a) find the value of k for which $\mathbf{M}$ has a repeated eigenvalue.

For that value of k,

b) find all the eigenvalues and corresponding eigenvector.

c) Find the trace($\mathbf{M}$)

Question 17.

$$\mathbf{M} = \begin{pmatrix} 3a-b-c & -a+b & -a+c \\ 3a-2b-c & -a+2b & -a+c \\ 3a-b-2c & -a+b & -a+2c \end{pmatrix}$$

where a, b and c are constants.
Giving your answer in terms of a, b and c:

a) Find $\det \mathbf{M}$
b) Find the eigenvalues of $\mathbf{M}$
c) Find the trace($\mathbf{M}$)

Question 18. Given that the matrix

$$\mathbf{M} = \frac{1}{2}\begin{pmatrix} a+c & a-c & c-a \\ a-b & a+b & b-a \\ c-b & b-c & b+c \end{pmatrix}$$

where a, b and c are constants.
Giving your answer in terms of a, b and c:

a) Find $\det \mathbf{M}$
b) Find the eigenvalues of $\mathbf{M}$
c) Find the trace($\mathbf{M}$)